

Job Site Feasibility Research

Archilles Job Pipeline — June 2026 | Prepared by Jade (Hermes CLI Agent)

1. Executive Summary

We evaluated 11 job sites for automated discovery by Archilles, a Hermes agent running on a Hetzner VPS. Each site was tested for login requirements, anti-bot defenses, API availability, and headless browser viability. The goal: identify which sites Archilles can reliably scrape without legal risk or technical headaches.

Core Strategy

Many UK tech companies use Lever, Ashby, or Greenhouse as their Applicant Tracking System (ATS). These ATS platforms expose clean, unauthenticated APIs. Instead of fighting LinkedIn or Indeed's bot defenses, Archilles will: (1) discover companies via open sources (Hacker News, reed.co.uk), then (2) go directly to their ATS APIs for structured job data. Skip the middlemen entirely.

2. Full Site Evaluation Table

Site	Verdict	Login Required	Anti-Bot	API Available	Method
Lever (jobs.lever.co)	WHITELIST	No	None	Yes — REST JSON	api.lever.co/v0/postings/{company}?mode=json
Ashby (jobs.ashbyhq.com)	WHITELIST	No	None	Yes — REST JSON	api.ashbyhq.com/posting-api/job-board/{company}
Greenhouse (boards.greenhouse.io)	WHITELIST	No	None	Clean HTML	HTML parse — no API key needed
HN "Who Is Hiring"	WHITELIST	No	Minimal (1 req/s)	Firebase + Algolia	Monthly thread, 400+ comments, keyword-filterable
reed.co.uk	GRAYLIST	No	None observed	No	Playwright scrape — clean HTML, 197 Python London jobs
cwjobs.co.uk	GRAYLIST	No	None observed	No	Playwright — 103 Python backend jobs in 14 days
cord.co	GRAYLIST	Partial	Minimal	No	UK tech focused, 13k+ positions, needs care
Workday (myworkdayjobs.com)	GRAYLIST	No*	Cookie wall	No	Heavy SPA, SSR content, Playwright with waits
LinkedIn (linkedin.com/jobs)	BLACKLIST	Yes	Aggressive	No (marketing only)	Legal risk, fingerprinting, hiQ precedent
Indeed (uk.indeed.com)	BLACKLIST	No*	Cloudflare	RSS blocked	VPS IP blocked on first request
Glassdoor	BLACKLIST	No*	Cloudflare	None	Instant "Humans only" challenge
Otta / Welcome to the Jungle	BLACKLIST	Yes	reCAPTCHA	None	Hard login wall before any jobs visible

3. Tier Breakdown

Tier 1 — Plug and Play (APIs, zero scraping)

These four sources require **no scraping at all**. They expose public APIs or clean HTML that Archilles can query with a single curl command. Zero anti-bot risk. Zero legal exposure.

Source	API Pattern	Example
Lever	GET /v0/postings/{company}?mode=json	Spotify: 166 jobs, many London-based
Ashby	GET /posting-api/job-board/{company}	Returns {"jobs": [...], "apiVersion": "1"}
Greenhouse	GET /{company} → HTML parse	GitLab: 157 jobs, cleanly rendered
HN Who Is Hiring	Firebase /v0/item/{id}.json	June 2026 thread: ID 48357725, 402 comments

Tier 2 — Scrape with Care

These UK-specific sources need Playwright-based scraping. They didn't block our test VPS, but they require rate limiting, user-agent rotation, and respectful delays. Used primarily for **company discovery** — find companies here, then switch to their Tier 1 ATS for clean job data.

reed.co.uk

cwjobs.co.uk

cord.co

Workday

Tier 3 — Not Worth It

LinkedIn (legal risk, login wall, aggressive anti-bot), Indeed (Cloudflare blocks VPS IPs outright), Glassdoor (Cloudflare challenge), and Otta (hard login wall with reCAPTCHA) are blacklisted. The cost of proxies, account management, and legal exposure outweighs any benefit — especially when Tier 1 already covers the same jobs via company ATS boards.

4. Why This Strategy Works

The key insight: **LinkedIn and Indeed are middlemen, not sources**. When a company posts a job, it ultimately lands on their ATS (Lever, Ashby, Greenhouse, or Workday). LinkedIn/Indeed just aggregate and wrap ads around it.

By discovering companies through open channels (HN monthly thread, reed.co.uk company names) and then hitting their ATS APIs directly, Archilles gets:

- **Structured data** — JSON with title, location, description, department, apply URL
- **No auth** — these ATS APIs are public by design (companies embed them on their careers pages)
- **No legal risk** — public APIs, no ToS circumvention, no scraping behind login
- **No proxy costs** — works from a plain Hetzner VPS with zero blocks

5. Recommended Deployment Schedule

Phase	Sources	Frequency	Effort
Phase 1 — Immediate	HN Who Is Hiring (Algolia API)	Monthly (1st of month)	1 cron job, ~20 lines of bash
Phase 2 — Week 1	Lever, Ashby, Greenhouse APIs — curated company list	Daily	1 cron job per ATS, company list in a text file
Phase 3 — Week 2	reed.co.uk, cwjobs.co.uk (company discovery → Tier 1)	Every 2 days	Playwright scripts, company name extraction
Phase 4 — Later	cord.co, Workday, findwork.dev	Weekly	Per-site Playwright scripts

6. Company Discovery Pipeline

How Archilles finds companies to query

Step 1: Scrape HN "Who Is Hiring" monthly thread → extract company names from "London", "Remote UK", "UK" comments.
Step 2: Scrape reed.co.uk search results → extract company names from listing cards.
Step 3: For each company name, check: do they use Lever? Ashby? Greenhouse?
Step 4: If yes → query their ATS API. If no → skip or add to Workday queue.
Step 5: Deduplicate across sources by company+title fingerprint.

7. Risk Mitigation

Risk	Mitigation
Hetzner IP gets rate-limited on reed.co.uk	2-day interval between scrapes, random delays, rotate user agents
Company stops using Lever/Ashby/Greenhouse	Graceful fallback — if API returns 404, mark company as "ATS changed"
HN thread format changes	Manual review of first monthly scrape; alert if comment count drops >50%
Job duplication across sources	Fingerprint by (company, title, location) triplet; deduplicate before evaluation
GDPR / legal exposure from scraping	Only scrape public, unauthenticated pages. No login circumvention. Respect robots.txt.

Note: LinkedIn and Indeed are blacklisted not because scraping is technically impossible, but because it's not worth it. The same jobs appear on company ATS boards (Tier 1) without any of the legal or technical overhead. For every hour we'd spend fighting Cloudflare and LinkedIn's legal team, we could have processed 200 jobs through clean APIs.

8. Verdict

4 whitelist sources (Lever, Ashby, Greenhouse, HN) provide clean, structured, legally-safe job data with zero scraping complexity. Combined with **4 graylist sources** (reed, cwjobs, cord, Workday) for company discovery, Archilles can cover the UK Python/backend job market comprehensively without ever touching LinkedIn or Indeed.

The total implementation for Phase 1+2 is approximately **4 cron jobs and 150 lines of bash** — no proxies, no captcha solvers, no account management. The smart play is spending engineering time on the evaluation pipeline and pack generation, not on fighting bot defenses.